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(54) Title: HOUSING ASSEMBLY FOR AN ELECTRIC MACHINE FOR DRIVING A MOTOR VEHICLE, AND ELECTRIC MACHINE WITH SUCH A HOUSING ASSEMBLY

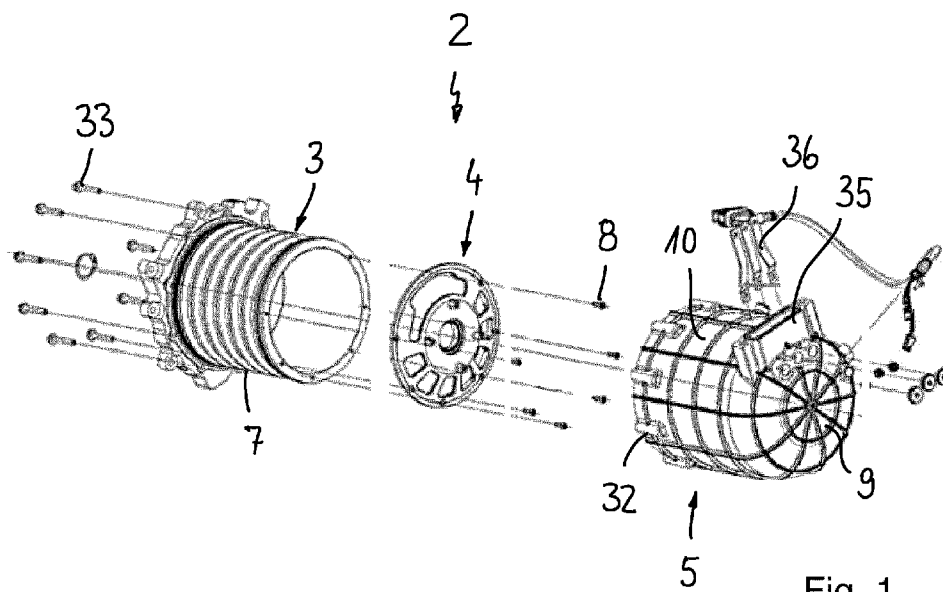


Fig. 1

(57) Abstract: The invention relates to a housing assembly for an electric machine for driving a motor vehicle, comprising: an inner housing part (3) for receiving a stator (43) of the electric machine, with a base portion (6) and an inner jacket portion (7) integrally formed of a metal material containing aluminum, wherein the inner jacket portion (7) comprises on an outer circumferential face a channel structure (12) for a cooling fluid; a side part (4) made of a metal material containing aluminum and being connected to an end of the inner housing part (3) opposite the base portion (6); an outer housing part (5) with a base portion (9) and an outer jacket portion (10) which are integrally formed of a plastic material; wherein the outer housing part (5) is mounted on the inner housing part (3) such that the outer jacket portion (10) covers the inner jacket portion (7), with the channel structure (12) being closed to form a circumferential cooling fluid channel (11). The invention further relates to an electric machine (42) with such a housing assembly.

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Housing assembly for an electric machine for driving a motor vehicle, and
electric machine with such a housing assembly

Description

The invention relates to a motor housing assembly for an electric machine for driving a motor vehicle, and an electric machine with such a motor housing assembly.

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From EP 3859951 A1, a motor housing assembly for cooling a motor assembly is known. The motor housing assembly includes a stator and a rotor. To reduce a weight and a manufacturing cost, the motor housing assembly includes an aluminum housing coupled to an outer circumferential surface of a motor assembly and configured to form a first space between the aluminum housing and the motor assembly, the first space including an oil flow channel, and a plastic housing coupled to an outer circumferential surface of the aluminum housing and configured to form a second space between the plastic housing and the aluminum housing, the second space including a water flow channel.

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From US 2017/0302133 A1, an electric pump motor is known comprising a permanent magnet rotor, a wound stator and a motor housing made of plastic material.

From US 2006/0071567 A1, a housing assembly is known for a permanent magnet motor. The housing assembly includes a ferrous flux ring structure. A plastic housing encapsulates the exterior of the flux ring structure. A bearing is provided in a closed end of the housing.

From US 2019 0027993 A1, an electric motor utilized in vehicle actuators is known including an electrically conductive plastic housing.

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From US 10 680 486 B2, an external rotor motor is known, including a rotary shaft, a plastic-packaged stator, a plastic-packaged body, and an external rotor sleeving on the plastic-packaged stator.

- 5 The present invention is based on the object to propose a housing assembly for an electric machine for driving a motor vehicle, which housing assembly provides low weight and contributes to energy efficiency and sustainability. A further object is to propose an electric machine with such a housing assembly that has a good efficiency.
- 10 According to the invention, a housing assembly for an electric machine for driving a motor vehicle is proposed, comprising: an inner housing part for receiving a stator of the electric machine, wherein the inner housing part comprises a base portion and an inner jacket portion which are integrally formed of a metal material that includes alumi-
- 15 num, wherein the inner jacket portion comprises on an outer circumferential face a channel structure for a cooling fluid; a side part made of a metal material that includes aluminum, wherein the side part is connected to an end of the inner housing part opposite to the base portion; and an outer housing part that comprises a base portion and an outer jacket portion which are integrally formed of a plastic material, wherein the outer housing part is mounted on the inner housing part such that the outer jacket
- 20 portion covers the inner jacket portion, with the channel structure being closed so as to form a circumferential cooling fluid channel.

An advantage of the proposed housing assembly is that the inner housing part and the side part, which are both formed of a metal material, jointly form a closed mechanical

25 structure thus providing a high stiffness, wherein at the same time the housing has a very low weight due to the outer housing part made of plastic material. Overall, the housing assembly contributes to a high energy efficiency and/or cost efficiency.

In the context of the present disclosure, the term outer housing part relates to the outer

30 jacket portion into which the inner housing part is inserted with its inner jacket portion. The outer housing can also be referred to as plastic housing part, and the inner housing part as metal housing part. Both, the outer and the inner housing parts are preferably pot-shaped, in other words, they include a base portion and a jacket portion made in one piece. The material of the inner housing part and the side part can be the same or

different from each other. The metal material of the inner housing part and the side part can have a higher stiffness than the plastic material of the outer housing part. The plastic material can have a lower density than the metal material.

- 5 The inner housing part can be made of pure aluminum or an aluminum alloy, which can comprise more than 90 % by weight of aluminum, for example, with further elements of together up to 10 weight % being optionally included. The outer housing part is made of a plastic material which can be a fibre reinforced plastic, for example. Preferred material properties of the plastic material are any type of high strength plastic or
10 plastic composite.

According to an embodiment, the base portion of the inner housing part can have a through-opening to receive a first bearing for rotatably supporting a rotor shaft of the electric machine, and the side part connected to the inner housing part can have a side
15 part opening to receive a second bearing for rotatably supporting the rotor shaft of the electric machine. Thus, in particular radial forces acting from the first and second bearings to the housing will be well supported, with dynamic and static forces efficiently accommodated by a closed force path due to the closed casing configuration of the metal inner housing part and the metal side part connected therewith. The axial forces
20 of the rotor shaft are preferably supported only by the bearing of the base portion of the inner housing part.

The base portion of the inner housing part serves as a bearing carrier for the rotor shaft. It can also be referred to as A-shield. A particular high stiffness may be achieved
25 in an embodiment where the base portion, on an outer side face thereof, comprises circumferentially distributed ribs that extend from an inner sleeve portion radially outwardly, in particular with a decreasing axial height.

A contribution to particularly high structural properties can be made in an embodiment,
30 where the base portion of the inner housing part and/or the side part comprise a reinforcement ring for receiving the respective first or second bearing. The reinforcement ring can be made of a metal material, in particular steel, that has a greater strength than the respective metal material of the base portion and the side part.

The side part is connected to the inner housing part in a rigid manner and serves as a bearing carrier for the rotor shaft. The side part can also be referred to as B-shield, bearing carrier, bearing fixation rim or simply rim. The connection can be achieved by axially mounting the side part to a side face of the first housing part, for example by circumferentially distributed connecting elements, such as screws.

The side part can be disc-shaped in one embodiment. As an alternative, the side part can also have an open structure to reduce weight. In particular, the side part can include an inner ring portion forming the opening for the shaft bearing, an outer ring portion or rim to be connected to the inner housing part and circumferentially distributed web portions that extend radially between the inner ring portion and the outer ring portion. In this case, axial through-openings are formed circumferentially between respective web portions which contributes to a reduced weight. The number of webs and through-openings can be, for example, between three and 15. The connecting elements for connecting the side part to the inner housing part can be arranged circumferentially in the region of the webs, without being limited thereto.

The webs and/or through-openings can have the same or different shapes and/or circumferential extensions. In an embodiment, one of the openings may have a greater circumferential extension so as to form a passage for electronic components, such as electric cables and/or connectors. The wall portion surrounding the opening can be used to attach one or more electronic components.

According to an embodiment, the base portion of the outer housing part is closed to form an isolated chamber together with the inner housing part. A space, respectively chamber may be formed between the closed base portion of the plastic housing part and a side face of the bearing carrier. This chamber can be configured to accommodate electronical and/or mechanical components of the electric machine, such as a resolver or electric connectors.

The plastic housing part can comprises a reinforcement ring at an end facing the base portion of the metal housing part. The reinforcement ring can comprise an inner circumferential face that is centered on an outer circumferential contact face of the metal

housing part, such that the plastic housing part is coaxially arranged on the metal housing part. The coaxial connection section between the inner circumferential face of the plastic housing part, respectively reinforcement ring, and the outer contact face of the metal housing part is preferably an interference fit connection. Other sections between the plastic housing part and the metal housing part, in particular in the region along the cooling channel, may have clearance. The reinforcement ring is preferably made of a material that has a higher strength than the plastic material of the main body of the plastic housing part. In particular, a metal ring can be used as reinforcement ring. The reinforcement ring can include a plurality of circumferentially distributed radial projections that can be configured as flange connections for being connected to the metal housing part.

A first sealing can be provided between an inner circumferential face of the reinforcement ring and a circumferential groove of the inner housing part. However, the sealing can also be provided so as to contact an inner circumferential face of the plastic housing part instead of a reinforcement ring. The sealing can be axially shorter than the axial length of the reinforcement ring. Thus, the reinforcement ring is radially supported on the outer contact face of the metal housing part axially adjacent to the first sealing.

The plastic housing part can have a plurality of circumferentially distributed connecting elements for being connected with the base portion of the metal housing part. The connecting elements can be configured as flange portions into which bolts may be screwed to attach the plastic housing to the metal housing. The dimensions of the plastic and metal housing parts can be such, that in the mounted condition, at least in one axial section, preferably close to the open end of the plastic housing part, the inner circumferential surface of the plastic housing part is in pretension contact with the outer circumferential surface of the metal housing part. In other axial sections, in particular along the axial extension of the cooling channel, there may be radial clearance between the inner face of the plastic housing and the outer face of the metal housing. At the axial end of the metal housing part, there may be an interference fit again between the metal and plastic housing part.

Furthermore, with respect to the above object, an electric machine for driving a motor vehicle is proposed, comprising: a housing assembly that can be configured according

to any one or more of the above embodiments; a stator connected in a rotationally fixed manner to the inner jacket portion of the inner housing part; and a rotor drivingly connected to a rotor shaft, with the rotor shaft being rotatably supported in the inner housing part and the side part about the rotor axis.

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The electric machine has the same advantages as mentioned with respect to the housing assembly so that reference is hereby made to the above description. The electric machine has a low weight, high stiffness and thus overall a high energy efficiency.

10 Preferred embodiments of the invention are described below using the drawing figures. Herein:

Figure 1 shows a housing assembly according to the invention in a 3D exploded view;

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Figure 2 shows the first housing part and side part of the housing assembly of Figure 1 in an assembled condition in a 3D sectional view;

Figure 3A shows the side part of the housing assembly of Figure 1 in a side view;

20 Figure 3B shows the side part of the housing assembly of Figure 1 in an axial view;

Figure 3C shows the side part of the housing assembly of Figure 1 in a sectional view according to 3C-3C of Figure 3B;

Figure 4A shows the second housing part of the housing assembly of Figure 1 in a 3D view from the bottom side;

25

Figure 4B shows the second housing part of the housing assembly of Figure 1 in a 3D view from the opening side;

Figure 5A shows an electric machine according to the invention in a longitudinal section with a housing assembly as shown in Figures 1 to 4;

30

Figure 5B shows the electric machine of Figure 5A in a 3D exploded view; and

Figure 6 shows a detail of the housing assembly of Figure 5A in a modified embodiment.

Figures 1 to 4B, which are described together below, show a housing assembly 2 according to the invention. The housing assembly 2 is for an electric machine for driving a motor vehicle (not shown). The housing assembly 2 includes an inner housing part 3 for receiving a stator of the electric machine, a side part 4 connectable to an end section of the inner housing part 3, and an outer housing part 5 that is connectable to the inner housing part 3.

The inner housing part 3 comprises a base portion 6 and a jacket portion 7 which are integrally formed of a metal material, in particular a light metal material such as aluminum or an aluminum alloy. To that extent, the inner housing part 3 can also be referred to as metal housing part. The metal housing part can be produced by a casting process, for example. The metal material can contain aluminum, for example, with an amount of at least 90 weight %, wherein further elements of together up to 10 weight % can optionally be included. The side part 4 can be connected to an end section of the inner housing part 3 opposite to the base portion 6 by suitable connecting means 8 such as screws, without being limited thereto. The side part 4 is also made of a metal material, in particular a light metal material such as aluminum or an aluminum alloy. The inner housing part 3 and the side part 4 can be made of the same or different metal materials.

The base portion 6 of the metal housing forms a bearing carrier for a rotor shaft of the electric machine, and can also be referred to as A-shield. A through-opening 16 is provided in the base portion 6 through which a rotor shaft of the electric machine can extend, wherein the rotor shaft can be rotatably supported by means of a first bearing 17. A clamping ring 18 can be provided to axially fix the bearing 17. The base portion 6 can optionally comprise, on an outer side face thereof, circumferentially distributed ribs 21 that extend from an inner sleeve portion 31 radially outwardly with a decreasing axial height.

At the opposite end, the side part 4 serves as a second bearing carrier for the rotor shaft, and can also be referred to as B-shield. The side part 4 comprises an opening 19 to receive a second bearing 20 for rotatably supporting the rotor shaft. The side part 4 can optionally comprise a reinforcement ring 47 for receiving the bearing 20. The

reinforcement ring 47 can be made of a metal material that has a greater strength than the metal of the side part 4.

Figure 2 shows the described unit comprising the inner housing part 3 with inserted first bearing 17 and side part 4 connected thereto with inserted second bearing 20 as a detail. Through this casing structure, forces acting from the first and second bearings 16, 20 to the metal housing and side part can be efficiently accommodated by a closed force path. The force path F is shown in Figure 5A. The bearing arrangement can be such that the first bearing 17 is configured as axially fixed bearing, whereas the second bearing at 20 can be configured as axially floating bearing.

The outer housing part 5 includes a base portion 9 and a jacket portion 10 which are integrally formed of a plastic material. To that extent, the outer housing part 5 can also be referred to as plastic housing part or simply plastic housing. The plastic material of the outer housing part can have a lower stiffness than the metal of the inner housing part 3 and/or side part 4. Furthermore, the plastic material can have a lower density than the metal material.

The plastic housing is mounted onto the metal housing such that the jacket portion 10 of the plastic housing covers the jacket portion 7 of the metal housing. The jacket portion 7 of the metal housing comprises, on an outer circumferential face thereof, a channel structure 12 with, for example, one or more circumferentially and/or helically extending channels formed by land portions 13. In the mounted condition, the channel structure 12 of the inner jacket portion 7 is closed by the outer jacket portion 10. The outer circumferential surface 14 of the land portions 13 can come into contact with the inner circumferential surface 15 of the outer jacket portion 10 and/or radial clearance may be provided. Thus, one or more closed cooling fluid channels 11 are formed between the inner and outer jacket portion 7, 10 through which a coolant can flow to cool the electric machine.

The side part 4, which is shown as a detail in Figures 3A to 3C, includes an inner ring portion 22 surrounding the cover opening 19 for the shaft bearing, an outer ring portion 23 to be connected to the metal housing part and circumferentially distributed web portions 24 that extend radially between the inner ring portion 22 and the outer ring

portion 23. Axial through-openings 25 are respectively formed circumferentially between adjacent web portions 24. In the present embodiment, the side part 4 comprises a plurality of openings 25 that have the same shape, and one opening 25' that has a greater circumferential extension of in particular more than 30° around the longitudinal axis A2. The greater opening 25' can serve to attach and/or pass through one or more electronic components, such as cables and connectors 30 for the stator of the electric machine (as shown in Figure 5A).

At the radially outer ring portion 23, the side part 4 comprises circumferentially distributed connecting elements 27 for rigidly attaching the side part 4 to the metal housing. Furthermore, as can be seen in Figures 3A and 3C, the side part 4 can include a circumferentially extending centering portion 31 that interacts with a respective centering face of the inner housing part 3 such that, in the mounted condition, the side part 4 is coaxially connected to the inner housing part 3. The side part 4 can comprise, at a side face of the inner ring portion 22, attaching means 26 to attach an electronic component, in particular a resolver for the electric machine. The resolver is configured as a rotary electronical transformer for measuring rotation of the electric machine.

Figures 4A and 4B show the outer housing part 5 as a detail, where the base portion 9 and the jacket portion 10 can be seen. The base portion 9 is closed, so that in the mounted condition, the plastic housing forms a liquid-tight casing for the electric machine. The plastic housing is formed such that, in the mounted condition (as shown in Figure 5A), a space respectively chamber 28 is formed between the closed base portion 9 of the plastic housing and the side part 4 connected to the metal housing. This chamber 28 provides room for an electronic component 29 connected to the side part 4, for example a resolver.

The outer housing part 5 includes a plurality of circumferentially distributed connecting elements 32 for being connected with the base portion 6 of the metal housing part. In the present embodiment, the connecting elements are configured as circumferentially distributed flange sections into which bolts 33 can be engaged for attaching the plastic housing to the base portion 6 of the metal housing. The base portion 6 has respective flange sections 34 through which the bolts can pass through to engage with the end portion of the plastic housing.

The outer housing part 5 further comprises a sealed connecting portion 35 for the bus bars 36 of the electric machine. The bus bars 36 can be overmolded into the connecting portion 35 of the plastic housing. Furthermore, the outer housing part 5 comprises
5 fluid connectors 37, 37' for the supply respectively return of the cooling channel 11 of the cooling fluid. For a high stiffness, the plastic housing can optionally comprise longitudinally and/or circumferentially extending ribs 38, 39.

A first sealing ring 40 is provided between an inner circumferential face of the outer
10 housing part 5 and a circumferential groove of the inner housing part 3. A second sealing ring 41 is provided at the axially opposite end between the inner circumferential face of the outer housing part 5 and a second circumferential groove of the inner housing part 3 (shown in Figures 2 and 5A).

Figures 5A and 5B show an electric machine 42 for driving a motor vehicle which includes a housing assembly according to the invention as described with respect to
15 Figures 1 to 4. The same and/or corresponding components are provided with the same reference signs as in Figures 1 to 4 so that reference is made to the above description with regard to the structure and mode of operation.

The electric machine 42 comprises, in addition to the housing assembly 2, a stator 43 connected in a rotationally fixed manner to the jacket portion 7 of the metal housing part and a rotor 44 drivingly connected to a rotor shaft 45. One end of the rotor shaft 45 is rotatably supported in the base portion 6 of the inner housing part 3 and the
20 opposite end of the rotor shaft 45 is rotatably supported in the side part 4. The shaft 45 is thus rotatably supported about the axis of rotation A2.

In a modified embodiment shown in Figure 6, the outer housing part 5 can comprise a reinforcement ring 46 at an end section thereof. The reinforcement ring 46 can be
30 made of a metal material, such as steel, for example. The reinforcement ring 46 comprises an inner circumferential face that is centered on the outer circumferential contact face 14' of the metal housing part so that the two parts (3, 5) are coaxially arranged. In an embodiment with reinforcement ring 46, the sealing 40 can be arranged within the axial extension of the reinforcement ring. In this case, the reinforcement ring 46 is

preferably axially longer than the sealing so that the reinforcement ring 46 is radially supported on the outer contact face 14' of the inner housing part 3 axially adjacent to the sealing ring 40.

Reference Numerals

2	housing assembly
3	inner housing part
4	side part
5	outer housing part
6	base portion (3)
7	jacket portion (3)
8	connecting means
9	base portion (5)
10	jacket portion (5)
11	cooling channel
12	channel structure
13	land portion
14, 14'	circumferential surface (13)
15	circumferential surface (9)
16	through-opening
17	bearing
18	clamping ring
19	cover opening
20	bearing
21	ribs
22	inner ring portion
23	outer ring portion
24	web portion
25	opening (4)
26	attaching means
27	connecting elements
28	chamber

29	electronic component
30	connector
31	sleeve portion
32	connecting elements
33	bolts
34	flange sections
35	connecting portion
36	bus bars
37, 37'	fluid connectors
38	ribs
39	ribs
40	sealing ring
41	sealing ring
42	electric machine
43	stator
44	rotor
45	rotor shaft
46	reinforcement ring
47	reinforcement ring
A	axis
F	force path

Claims

1. Housing assembly for an electric machine for driving a motor vehicle, comprising:

an inner housing part (3) for receiving a stator (43) of the electric machine, wherein the inner housing part (3) comprises a base portion (6) and an inner jacket portion (7) which are integrally formed of a metal material that includes aluminum, wherein the inner jacket portion (7) comprises on an outer circumferential face a channel structure (12) for a cooling fluid,

a side part (4) made of a metal material that includes aluminum, wherein the side part (4) is connected to an end of the inner housing part (3) opposite to the base portion (6),

an outer housing part (5) that comprises a base portion (9) and an outer jacket portion (10) which are integrally formed of a plastic material,

wherein the outer housing part (5) is mounted on the inner housing part (3) such that the outer jacket portion (10) covers the inner jacket portion (7), with the channel structure (12) being closed so as to form a circumferential cooling fluid channel (11).
2. Housing assembly according to claim 1,
characterised in that the metal material of the inner housing part (3) and the metal material of the side part (4) respectively have a greater stiffness and higher density than the plastic material of the outer housing part (5).

3. Housing assembly according to claim 1 or 2,
characterised in
that the base portion (6) of the inner housing part (3) has a through-opening (16)
to receive a first bearing (17) for rotatably supporting a rotor shaft (45) of the
electric machine (42), and
that the side part (4) connected to the inner housing part (3) has an opening (25)
to receive a second bearing (20) for rotatably supporting the rotor shaft (45) of
the electric machine.
4. Housing assembly according to any one of claims 1 to 3,
characterised in
that at least one of the base portion (6) of the inner housing part (3) and the side
part (4) connected thereto comprises a reinforcement ring (47) for receiving the
respective first or second bearing (17, 20),
wherein the reinforcement ring (47) is made of a metal material, in particular
steel, that has a greater strength than the metal materials of the base portion (6)
and the side part (4).
5. Housing assembly according to any one of claims 1 to 4,
characterised in
that the side part (4) includes an inner ring portion (22) forming the opening (25),
an outer ring portion (23) connected to the inner housing part (3) and circumfer-
entially distributed web portions (24) that extend radially between the inner ring
portion (22) and the outer ring portion (23).
6. Housing assembly according to any one of claims 1 to 5,
characterised in
that the side part (4) is connected to a side face of the inner housing part (3) by
circumferentially arranged connecting means (8).
7. Housing assembly according to any one of claims 1 to 6,
characterised in

that the base portion (6) of the inner housing part (3), on an outer side face thereof, comprises circumferentially distributed ribs (21) that extend radially outwardly from an inner sleeve portion (31).

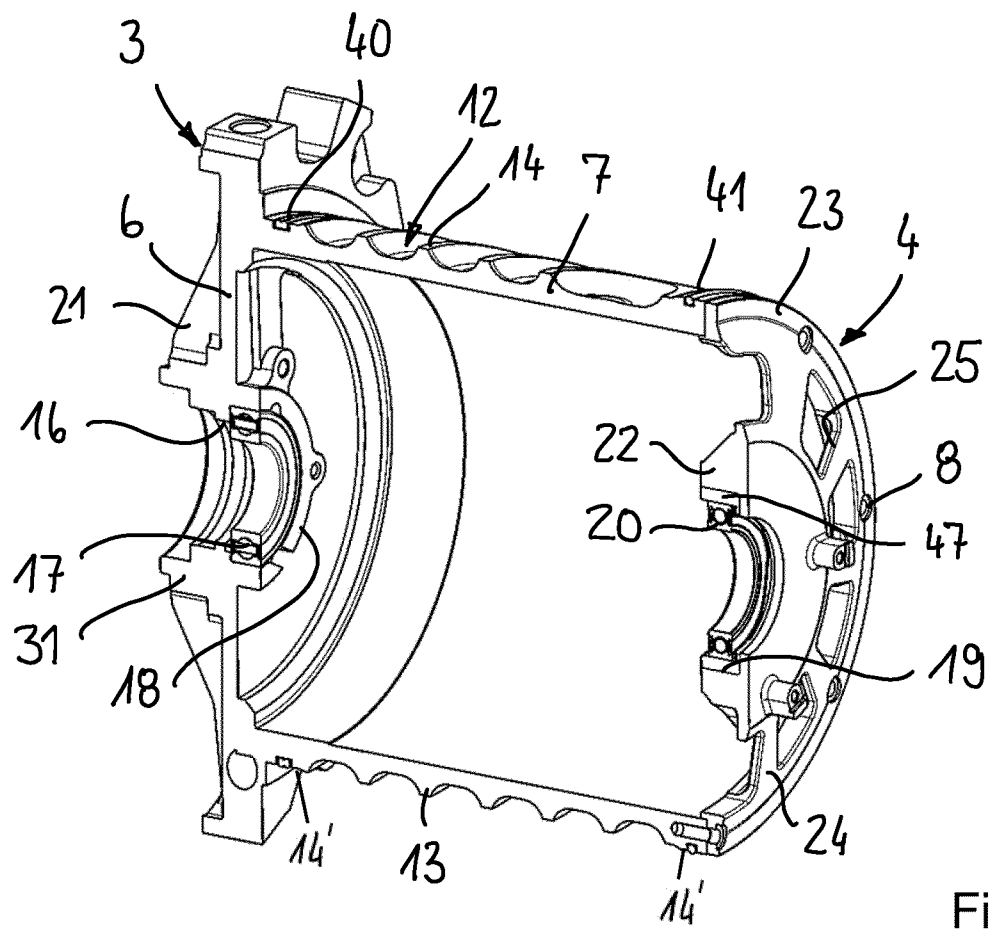
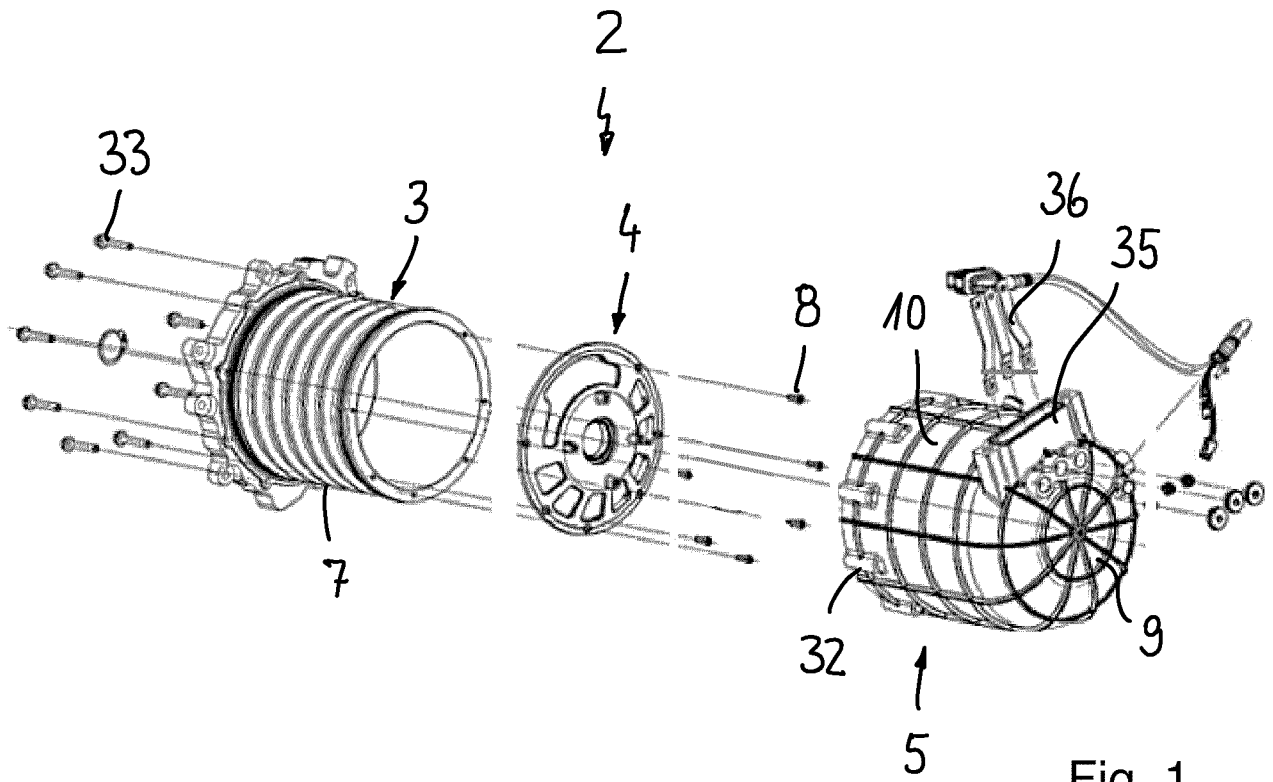
8. Housing assembly according to any one of claims 1 to 7, characterised in
that the base portion (9) of the outer housing part (5) is closed, wherein a space (28) is formed between the base portion (9) of the outer housing part (5) and a side face of the side part (4).
9. Housing assembly according to any one of claims 1 to 8, characterised in
that the outer housing part (5), at an end facing the base portion (6) of the inner housing part (3), comprises an reinforcement ring (46), wherein the reinforcement ring (46) comprises an inner circumferential face that is centered on a circumferential contact face (14') of the inner housing part (3), such that the outer housing part (5) is coaxially arranged on the inner housing part (3).
10. Housing assembly according to claim 9, characterised in
that the reinforcement ring (46) is made of a material that has a higher strength than the plastic material of the main body of the outer housing part (5).
11. Housing assembly according to claim 9 or 10, characterised in
that the reinforcement ring (46) includes a plurality of circumferentially distributed radial projections that are configured as flange connections.
12. Housing assembly according to any one of claims 9 to 11, characterised in
that a first sealing (40) is provided between an inner circumferential face of the reinforcement ring (46) and a circumferential groove of the inner housing part (3).

13. Housing assembly according to claim 12,
characterised in
that the first sealing (40) is axially shorter than the reinforcement ring (46) so that the reinforcement ring is, axially adjacent to the first sealing (40), radially supported on the circumferential contact face (14') of the inner housing part (3).
14. Housing assembly according to any one of claims 1 to 13,
characterised in
that the outer housing part (5) has a plurality of circumferentially distributed connecting elements (32) for being connected with the base portion (6) of the inner housing part (3).
15. Electric machine for driving a motor vehicle, comprising:

a housing assembly (2) according to any one of claims 1 to 14,

a stator (43) connected in a rotationally fixed manner to the inner jacket portion (7) of the inner housing part (3), and

a rotor (44) drivingly connected to a rotor shaft (45), with the rotor shaft (45) being rotatably supported in the inner housing part (3) and the side part (4) about the rotor axis (A2).



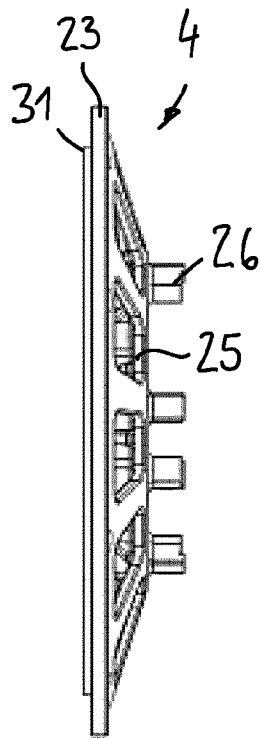


Fig. 3A

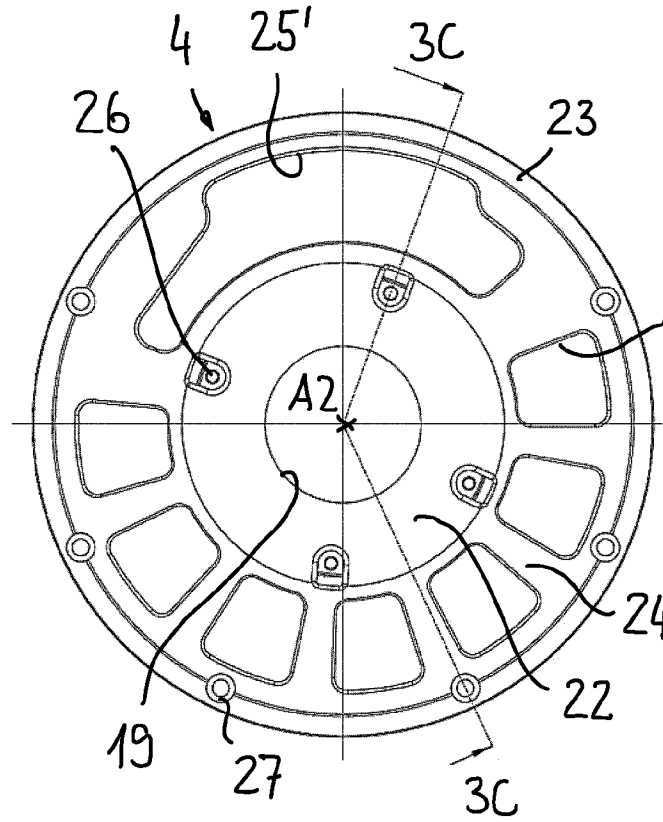


Fig. 3B

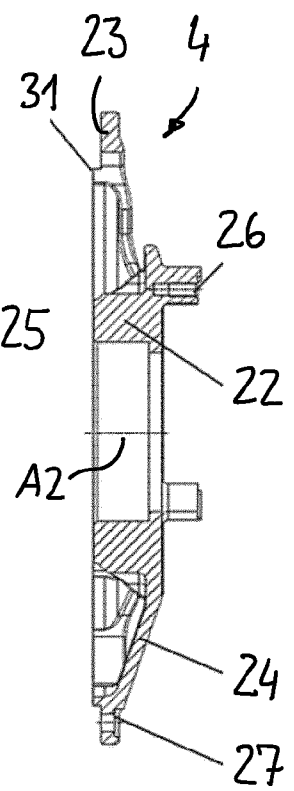


Fig. 3C

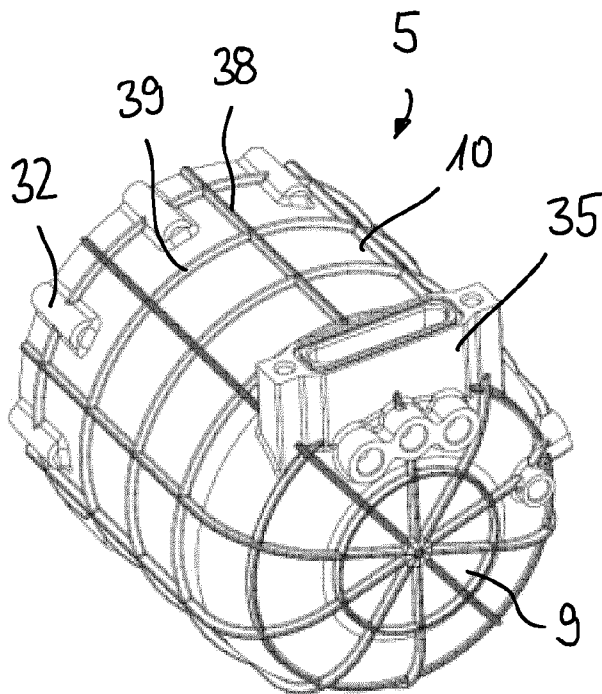


Fig. 4A

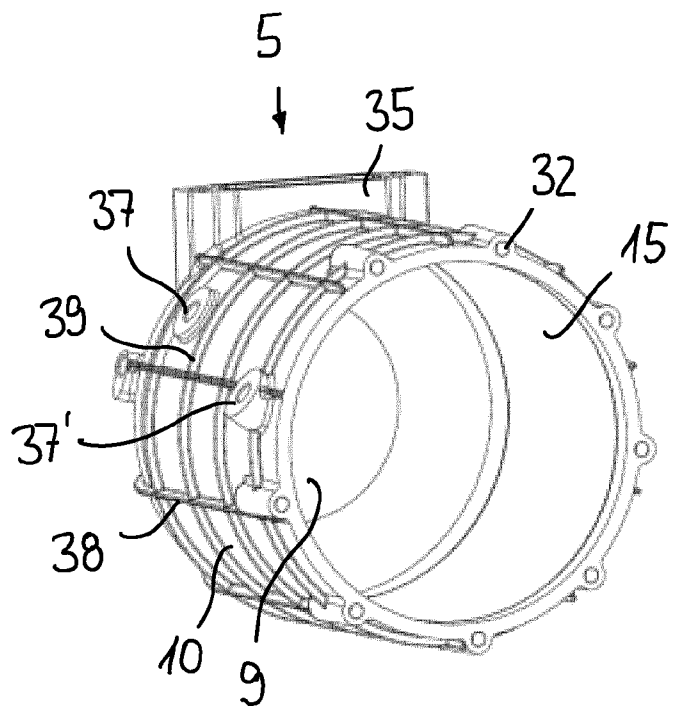


Fig. 4B

Fig. 5A

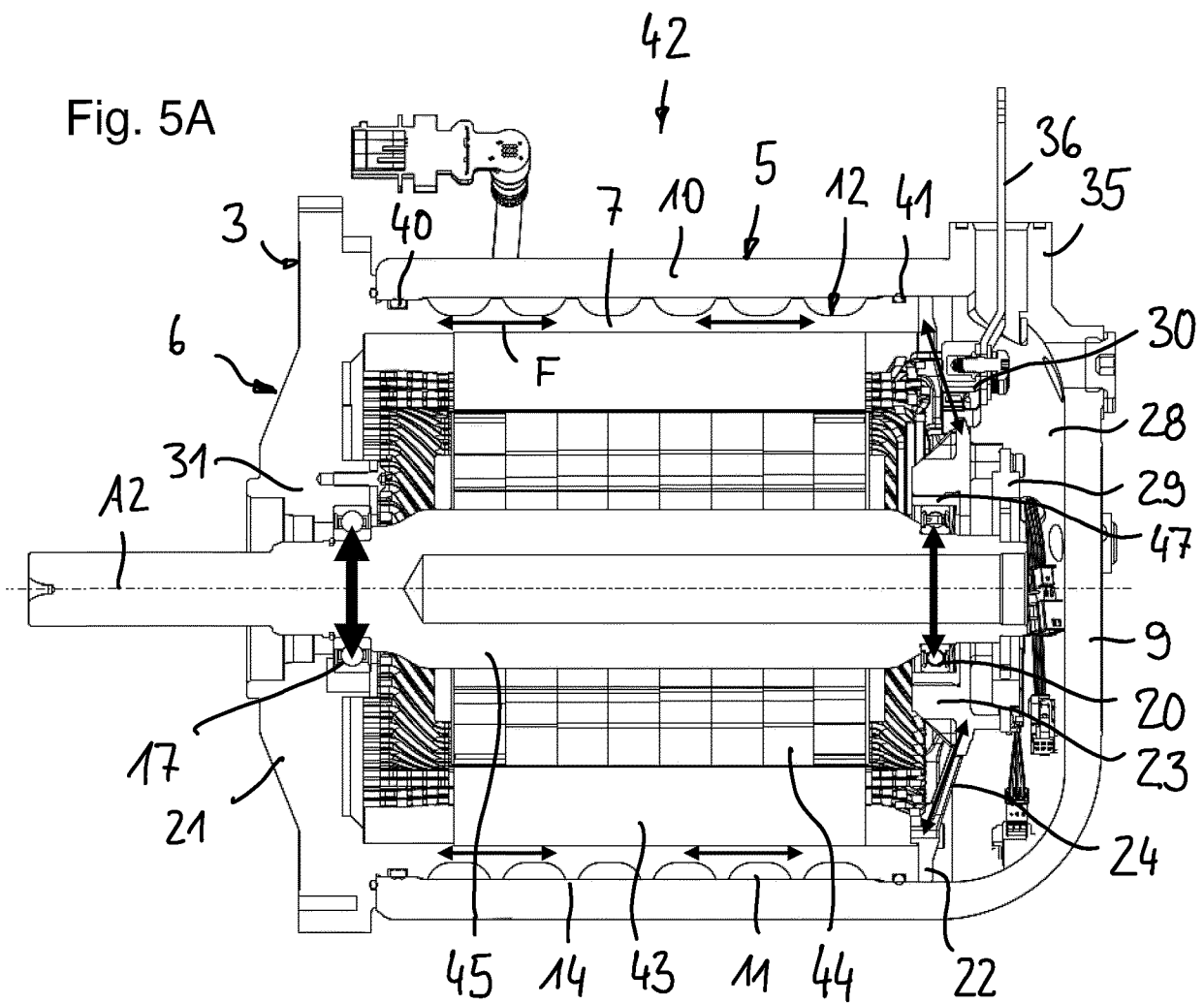
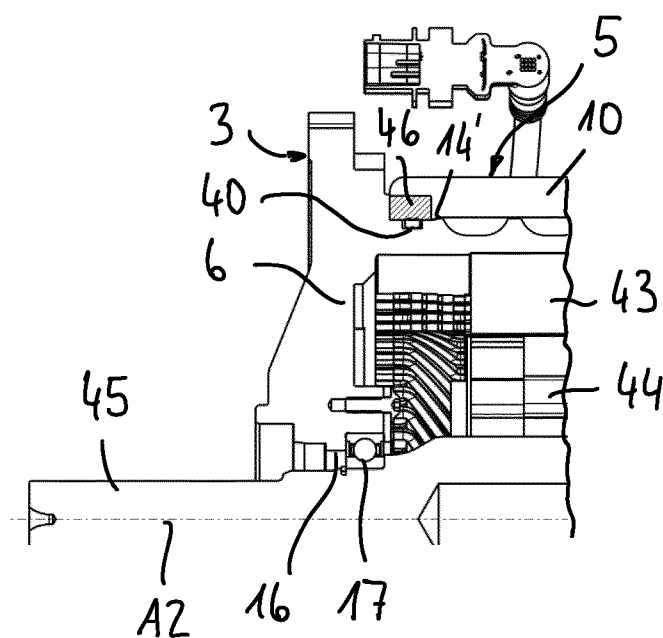


Fig. 6



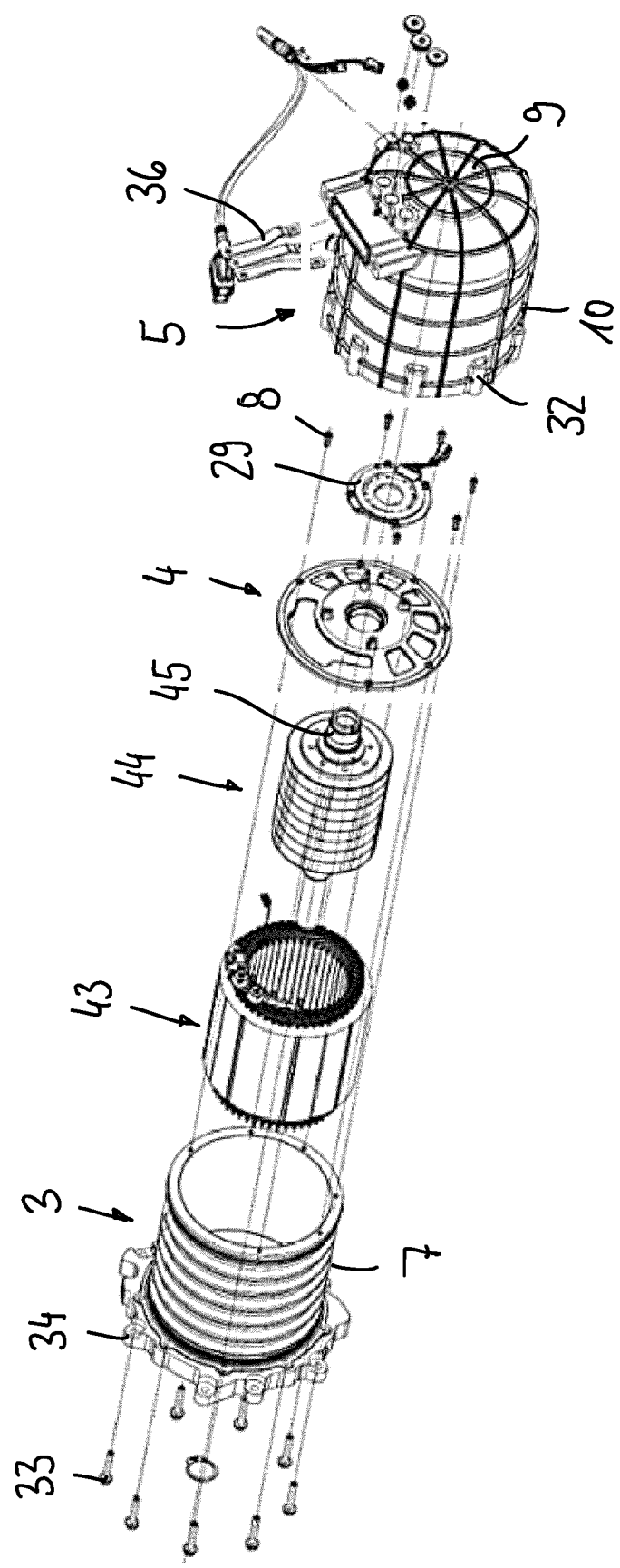


Fig. 5B

INTERNATIONAL SEARCH REPORT

International application No
PCT/EP2022/077202

A. CLASSIFICATION OF SUBJECT MATTER INV. H02K5/173 H02K5/20 H02K5/22 ADD. H02K5/06 H02K5/08		
According to International Patent Classification (IPC) or to both national classification and IPC		
B. FIELDS SEARCHED Minimum documentation searched (classification system followed by classification symbols) H02K		
Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched		
Electronic data base consulted during the international search (name of data base and, where practicable, search terms used) EPO-Internal		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	FR 3 108 003 A1 (NOVARES FRANCE [FR]) 10 September 2021 (2021-09-10) figures 1,2	1-11, 14, 15

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☒ Further documents are listed in the continuation of Box C. ☒ See patent family annex.		
* Special categories of cited documents : "A" document defining the general state of the art which is not considered to be of particular relevance "E" earlier application or patent but published on or after the international filing date "L" document which may throw doubts on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified) "O" document referring to an oral disclosure, use, exhibition or other means "P" document published prior to the international filing date but later than the priority date claimed "T" later document published after the international filing date or priority date and not in conflict with the application but cited to understand the principle or theory underlying the invention "X" document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone "Y" document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art "&" document member of the same patent family		
Date of the actual completion of the international search		Date of mailing of the international search report
11 May 2023		22/05/2023
Name and mailing address of the ISA/ European Patent Office, P.B. 5818 Patentlaan 2 NL - 2280 HV Rijswijk Tel. (+31-70) 340-2040, Fax: (+31-70) 340-3016		Authorized officer Le Chenadec, Hervé

INTERNATIONAL SEARCH REPORT

International application No

PCT/EP2022/077202

C(Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		
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